

The Hong Kong Polytechnic University

Subject Code	ME42013
Subject Title	Robot Mechanisms and Motion
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME23001 Engineering Mechanics AND Students should have basic knowledge of Dynamics.
Objectives	This course covers topics including classifications of mechanisms; velocity, acceleration and force analysis (e.g. for linkages, cranks, sliders, and cams); balancing of rotating and reciprocating machinery; gears and gear-trains; graphical and computer-oriented methods of analysis for mechanisms; applications of different mechanisms in mechanical systems (e.g., engines, manufacturing systems).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Recognize concepts and applications of mechanisms (e.g., cam, flywheels, gear and gear trains, etc.) in mechanical systems. b) Conduct graphical and analytical analyses and/or use computer programs to obtain position, velocity, and acceleration of a given mechanism. c) Apply graphical and analytical methods to obtain static, inertia, and dynamic forces on planar mechanisms. d) Analyze motions and solve for forces in simple, compound, and planetary gear trains. e) Design four-bar and/or slider crank mechanisms for function generation or rigid body guidance.
Subject Synopsis/ Indicative Syllabus	This is a course that develops the knowledge and skills mechanical engineers need to design and analyze mechanisms in mechanical systems. The gained knowledge and experiences in this course will be essential for any future project activities. Introduction to Mechanisms and Kinematics • Types of mechanisms, Definitions and terminologies, Kinematic Chains and Pairs, Mechanism Mobility and Inversion. Mechanics of Rigid Bodies • Relative velocity and acceleration, Limit positions and time ratio of mechanisms, Transmission angle, Instantaneous centre of velocity, Kennedy's theorem, Kinetics, Equations of equilibrium. Graphical Kinematics Analysis • Velocity analysis using instantaneous centres, Velocity polygon analysis, Velocity image, Acceleration polygon analysis, Acceleration image. Analytical Kinematics • Loop closure equation, Complex vector analysis with one and multiple loops. Graphical Force Analysis

	- Two and three force member, Force transmission, Force polygons, Static force analysis, principle of superposition, Dynamic force analysis. Analytical Force and Balancing - Force analysis (e.g., four-bar mechanism, slider crank mechanism), Unbalance and balancing, Force balancing of mechanisms. Flywheels and gyroscopic forces - Mathematical formulation for flywheels, Branched systems, Gyroscopic effects. Cam design - Nomenclature, Pressure angle, Displacement diagram, Follower motions, Disc cam profile, Positive motion cams, Program cam design Gears and gear train - Fundamental law of gearing, Involute and Cycloidal tooth gearing, Nonstandard spur, bevel, helical, and work gearing, Production of gears, Gear trains: ordinary gear trains, planetary gear trains, harmonic drives Synthesis of mechanism - Design of four-bar and slider crank mechanisms: function generation and rigid body guidance.																													
Teaching/Learning Methodology	- Lectures aim at providing students with fundamental knowledge required for understanding and analysing different networked systems, including its mathematical models, controller design, and algorithms. (Outcomes a to e) - Tutorials aim at enhancing students’ analytical and problem-solving skills in networked robots and IoT. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to e) - The experiments/project aim to provide hands-on experience for developing networked robots and IoT and reinforcing the acquired knowledge. (Outcomes a to e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Project report and presentation</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes					a	b	c	d	e	1. Lectures	√	√	√	√	√	2. Tutorials		√	√	√	√	3. Project report and presentation	√	√	√	√	√
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignments</td><td>15 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Quizzes</td><td>20 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Group Project</td><td>20 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Examination</td><td>45 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table>		Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Assignments	15 %	√	√	√	√	√	2. Quizzes	20 %	√	√	√	√	√	3. Group Project	20 %	√	√	√	√	√	4. Examination	45 %	√	√	√	√	√	Total	100 %										
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:																																																					
	Overall Assessment: $0.45 \times \text{Examination} + 0.55 \times \text{Continuous Assessment}$																																																					
1. The continuous assessment will comprise three components: team project (20%), quizzes (20%) and homework (15%). The team project, quizzes and homework are aimed at evaluating their understandings on robotic mechanisms and motion systems and enhancing the integration of their knowledge learnt.																																																						
2. The examination (45%) will be used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently, and to determine the degree of achieving the subject learning outcomes.																																																						
Student Study Effort Expected	Class contact:																																																					
	▪ Lecture					30 Hrs.																																																
	▪ Tutorial/Presentation					9 Hrs.																																																
	Other student study effort:																																																					
	▪ Reading and review					36 Hrs.																																																
	▪ Coursework (assignments, project)					40 Hrs.																																																
	Total student study effort					115 Hrs.																																																
Reading List and References	<u>Required Course Materials</u> Recommended Textbook: Mechanics of Machines, 2nd Edition, William Cleghorn & Nikolai Dechev, Oxford University Press, 2014, ISBN: 978-0195384086																																																					
	<u>Optional Course Materials</u> 1. Theory Of Machines and Mechanisms, 4th Edition, Gordon R. Pennock & Joseph E. Shigley John J. Uicker, Oxford University Press, 2010, ISBN: 978-0195371239 2. Machines and Mechanisms: Applied Kinematic Analysis, 4th Edition, David H. Myszka, Prentice Hall, 2012, ISBN: 978-0132157803 3. Design Simulation Technologies ® Working Model-2D Guide https://www.design-simulation.com/documents/WM/wmdemotoolkitpro.pdf 4. MathWorks® Matlab User Guide http://www.mathworks.com/help/pdf_doc/matlab/getstart.pdf																																																					
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